

Leveraging Limited Experimental Data with Machine Learning: Differentiating a Methyl from an Ethyl Group in the Corey–Bakshi–Shibata Reduction

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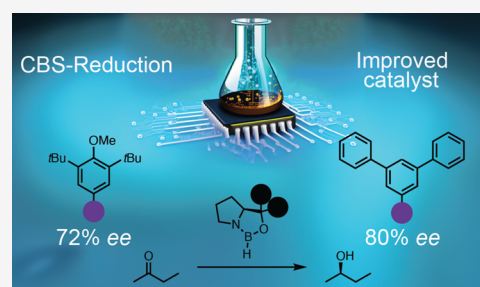


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ABSTRACT: We present a case study on how to improve an existing metal-free catalyst for a particularly difficult reaction, namely, the Corey–Bakshi–Shibata (CBS) reduction of butanone, which constitutes the classic and prototypical challenge of being able to differentiate a methyl from an ethyl group. As there are no known strategies on how to address this challenge, we leveraged the power of machine learning by constructing a realistic (for a typical laboratory) small, albeit high-quality, data set of about 100 reactions (run in triplicate) that we used to train a model in combination with a key-intermediate graph (of substrate and catalyst) to predict the differences in Gibbs activation energies $\Delta\Delta G^\ddagger$ of the enantiomeric reaction paths. With the help of this model, we were able to select and subsequently screen a small selection of catalysts and increase the selectivity for the CBS reduction of butanone to 80% enantiomeric excess (ee), the highest possible value achieved to date for this substrate with a metal-free catalyst, thereby also exceeding the best available enzymatic systems (64% ee) and the selectivity with Corey's original catalyst (60% ee). This translates into a >50% improvement in relative $\Delta G^\ddagger$ from 0.9 to 1.4 kcal mol⁻¹. We underscore the transformative potential of machine learning in accelerating catalyst design because we rely on a manageable small data set and a key-intermediate graph representing a combination of catalyst and substrate graphs in lieu of a transition-state model. Our results highlight the synergy of synthetic chemistry and data-centric approaches and provide a blueprint for future catalyst optimization.



INTRODUCTION

The differentiation of a methyl versus an ethyl group, for instance, in the metal-free catalytic reduction of butanone with a hydride source, is a long-standing and formidable challenge because the *re* and *si* faces of the substrate differ by just one methylene unit. Understanding how this challenge can be met will help develop catalysts for similarly challenging situations. One common reaction for this particular transformation is the Corey–Bakshi–Shibata (CBS)^{1,2} reduction, which employs a chiral boron-based catalyst, specifically an oxazaborolidine derived from L-proline or other chiral amines. This catalyst coordinates to borane (BH₃), the reducing agent, and together, they facilitate the reduction of the ketone in an enantioselective manner. The CBS reduction of prochiral ketones has been widely employed in organic syntheses due to their generally high enantioselectivity and broad substrate scope using, e.g., lactones, alkaloids, and steroids.^{3–5} For instance, it has been employed in the preparation of chiral isoxazole carbinols,⁶ the synthesis of (*R*)-phenylephrine,⁷ as well as the enantioselective syntheses of massadine,⁸ FR901464, and spliceostatin A.⁹

The lessons learned from such a challenging transformation could potentially be applied to other instances. The root cause of the “methyl/ethyl problem”—or generally two alkyl moieties—lies in the very subtle differences in their stereo-

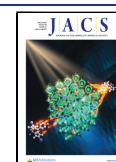
electronic properties. In terms of catalyst design, this leaves very few options other than steric size differences (as judged by van der Waals surfaces) utilized, e.g., in “confined” catalysts¹⁰ or London dispersion (LD) interactions.^{11–13} It has been demonstrated in the past few years that LD interactions indeed can be a decisive element in catalyst design^{1,14–17} by employing dispersion energy donors (DEDs),^{13,18} but would these be sufficient to meet the methyl/ethyl challenge? If so, which DEDs are the most suitable, and how can they be identified without having to generate a large catalyst library? Here, we address this issue using a small but typical catalyst library that forms the basis for a machine learning (ML) approach that suggests potentially better-performing catalysts. We demonstrate the efficacy of our data-centric approach with the CBS reduction of several prochiral ketones, and, in particular, for butanone, for which we achieved 80% enantiomeric excess (ee), which is by far the highest selectivity for this substrate reported to date for a metal-free system;

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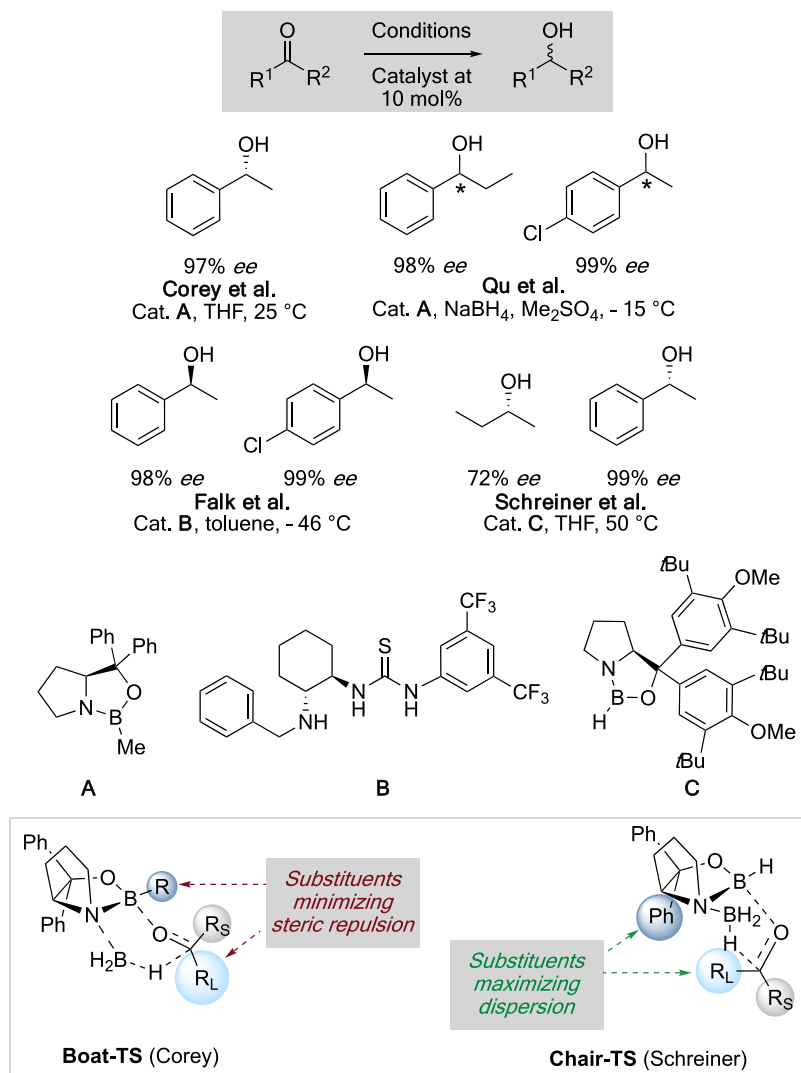
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Scheme 1. Selection of CBS Reductions for the Best (Aryl/Alkyl) and Most Challenging (Butanone) Substrates



fermenting yeast reached a selectivity of 64% ee.¹⁹ Transition-metal complexes on the other hand, such as the rhenium-based chiral Lewis acid $[(\eta^5-C_5H_5)Re(NO)(PPh_3)]^+$, gave an ee of 95%.²⁰

The best results for the CBS reduction of butanone were limited to 60% ee with the original Corey catalyst.^{1,21} The use of DED-equipped catalysts improved the ee to 72% after employing a laborious approach using detailed descriptors employing computational chemistry methods.¹

The CBS reduction (and many other additions to carbonyls) works well for prochiral ketones that bear stereoelectronically very different groups (Scheme 1), but it remains challenging to achieve high selectivity when the substituents are structurally very similar. As a consequence, the best results are obtained with aryl-alkyl groups, with acetophenone being the prime example that gives 97% ee in 99% yield.³ This has led to a classic selectivity model that builds on the concept of “steric hindrance” alone, placing the large substituents on the substrate and the boron atom on opposing sides of a boat-like transition state (Scheme 1).^{3,21,22} We demonstrated recently that such a model is highly insufficient by showing that the substituent at boron is largely inactive in the catalytic process and that an optimal balance between steric hindrance

and LD attraction must be met for catalyst optimization in a more favorable chairlike transition structure (Scheme 1).¹ How to strike this balance is, however, anything but trivial. Hence, we set out to use ML approaches, even though they are agnostic to molecular interactions, which can be utilized to move forward with catalyst design where our human ingenuity does not offer obvious choices. The CBS reduction was chosen as a model reaction here since it does react quite sensitively to DEDs in both the catalyst and the substrate and because it is a widely used reaction of high utility.

While design of experiment (DoE) and other types of optimizations can indeed help determine the optimal experimental conditions,^{23–26} these techniques do not address the inherent complexity of finding the optimal catalyst structure. Many ML approaches have emerged to tackle catalysis design but often require large experimental or computed data sets,^{27–47} a requirement that is often not met in prototypical experimental laboratories. Among these methods, the use of density functional theory (DFT) to compute catalyst properties and build linear correlations has been successfully tailored by Sigman and co-workers toward a comprehensive discovery platform.⁴⁸ Another approach, developed by Wheeler and colleagues, employs automated

transition-state calculations and correlations.⁴⁹ Corminboeuf and co-workers have also proposed the use of automated models based on volcano plots of reaction energetics.⁵⁰

Drawing from an array of ML approaches to innovate and optimize catalysts, we use ML to address the methyl/ethyl challenge and thus offer a complementary alternative to DoE and dedicated computational approaches. Our approach boils down to constructing a model that works solely based on the experimentally available compounds, without the need for additional in silico data generation, hence saving many thousands of CPU (and human effort) hours. As a result, we demonstrate the improvement of the efficiency and selectivity of the CBS reduction of butanone as a substrate by the selective preparation of ML-prioritized catalysts and compare our approach to alternatives, such as DFT descriptors, other molecular descriptors, different molecular representations, as well as other model architectures. To investigate the reasoning behind the final model, feature weights were visualized to confirm that the model had learned meaningful patterns.

RESULTS AND DISCUSSION

To ensure the reliability of ML approaches, high-quality data sets are a precondition. Since data extracted from the literature proved to vary vastly in quality and experimental conditions, we conducted 100 reactions (ten different catalysts and substrates) in triplicate with previously known catalysts (cf. [Supporting Information](#)). This process enabled us to construct a small but robust and balanced database, providing a foundation for implementing our ML strategy. Our initial database is based on the reactions that our group performed earlier and is depicted in [Scheme 2](#).¹

We have applied our understanding of the CBS reduction process,¹ focusing particularly on the critical step before the

reaction's rate-determining step. To visualize this concept, we developed what we call a "key-intermediate graph". This idea combines the structures of both the substrate and the catalyst into a single, connected graphical representation, as illustrated in [Figure 1](#). This approach allows us to examine the interaction

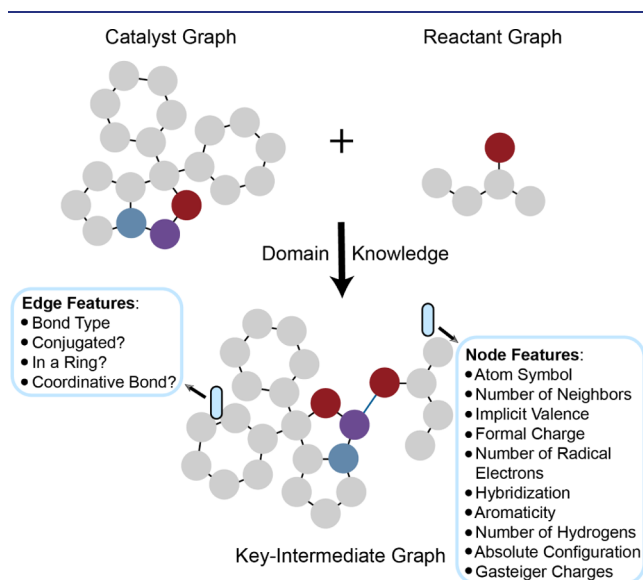


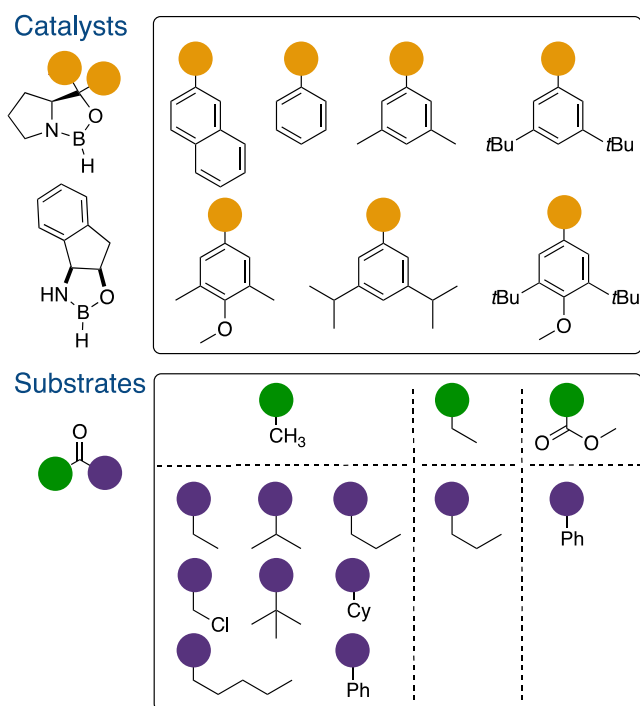
Figure 1. Combining the reactant and the catalyst graph to create a graph representing the CBS reduction's key-intermediate graph. The combination of both molecules occurs between the boron of the catalyst and the oxygen of the carbonyl. The nodes are colored based on the atom type: gray, carbon; purple, boron; blue, nitrogen; and red, oxygen.

between a substrate and a catalyst in a unified manner, simplifying the complex dynamics at play. Note that this approach does not entail the computation of any transition-state structures and thus remains agnostic with regard to the specific mechanism of the reaction.

This approach is easy to understand for the practicing chemist because we can readily envision what intermediates or transition structures may be specifically relevant for a particular reaction. As done often, this informed choice of a graph is based on nodes and edges, like a molecule, where atoms are represented by nodes and bonds by edges. The process of enhancing the mechanistically agnostic ML model with chemical knowledge is illustrated in [Figure 1](#). The combined graph is connected at the boron atom of the catalyst and the carbonyl oxygen atom in the substrate, which undergoes reduction during the CBS reaction. Each node and edge in the graph are assigned certain molecular properties, which are listed in [Figure 1](#). This graph thus offers a comprehensive representation of an essential structure along the CBS reaction path, providing a more information-rich input for our graph neural network (GNN)-based model compared to two disjoint graphs, which we initially tested. This key-intermediate graph representation proved beneficial for our task compared to the disconnected graph approaches, where the catalyst and the substrate are treated as individual graphs ([Figure 8](#)).

Utilizing our key-intermediate graph, we trained a GNN in a supervised manner to predict the Gibbs activation energy differences $\Delta\Delta G^\ddagger$ of the reaction paths leading to either enantiomer.⁵¹ We explored readily commercially available aryl bromo compounds, which were attached to the L-proline

Scheme 2. Overview of Our Initial Database^a



^aThe top shows the L-proline and indole-based catalysts, while the substrate scope is shown at the bottom.

scaffold via a Grignard reaction. We used our trained model to determine the optimal choice based on the predicted $\Delta\Delta G^\ddagger$ values. Remarkably, this approach immediately resulted in a catalyst that improved the ee from 72 to 80% for the CBS reduction of butanone. The time-saving benefits provided by this approach are depicted schematically in Figure 2. Using ML

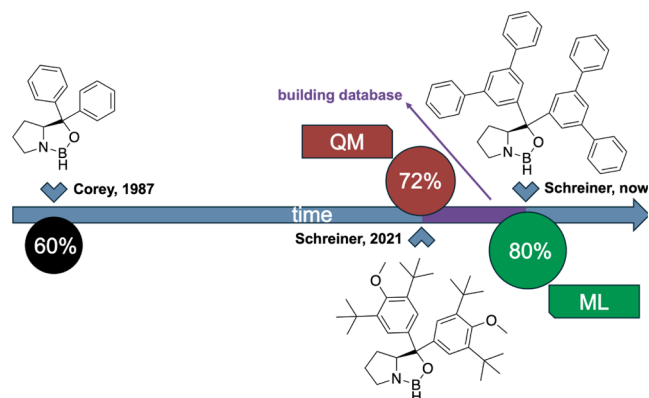


Figure 2. Schematic representation of the discovery timeline for new CBS catalysts using ML. QM indicates quantum mechanical computations. The ee is given in % in the circles.

to prioritize a small selection of suggested catalysts for testing potentially saves months of experimental work. Once a well-distributed data set (vide infra) is available, a better catalyst may essentially be predicted immediately.

While time savings are possible with high-quality, widely distributed data, as shown in Figure 3, this requires a combination of synthetic chemists and data-centric approaches to generate robust data to feed and optimize the algorithm effectively. To overcome the long-standing issue of publication bias,⁵² it is paramount to validate each reaction yield or selectivity metric multiple times. The publication bias describes the statistical trend toward “positive” rather than “negative” results in the literature, hence resulting in an artificially shifted database toward “positive” results. Figure 3 highlights a database highly suitable for ML by including “negative results” that may be lost due to publication bias in the literature. We show that by using such high-quality data, we can utilize the

power of machine learning via GNNs, even with only about 100 data points.

Models and Methods. Ensuring that our model enables accurate predictions necessitated careful consideration of the purity of catalysts, reducing agents, and substrates, as well as conducting multiple runs for each reduction. We implemented this rigor to ensure the integrity of the experimental data and to boost their trustworthiness. To achieve a balanced stoichiometry, we used 0.1 equiv of the precursor and 1.1 equiv of the reductant, i.e., borane dimethylsulfide, for catalyst formation. The rate of substrate addition proved critical since the fast addition of a substrate could compromise enantioselectivity because of an uncatalyzed background reaction. To ascertain the optimal conditions (illustrated in Figure 4), we varied the addition rate and found that an injection speed of $\leq 0.033 \text{ mL min}^{-1}$ is needed for consistent results. Additionally, we conducted temperature trials to identify the optimal temperature for catalyst-substrate reactivity. As illustrated in Scheme 1, when performing reductions with the CBS catalyst on ketones, the ideal reaction temperature can vary based on the conditions and catalysts. However, it is known that temperatures below ambient tend to diminish selectivity.^{53,54} Therefore, we opted for a temperature of 50 °C for all subsequent reactions, which was shown to be optimal previously and was also confirmed in our condition screening;⁵⁴ for the rate of addition, we chose $0.017 \text{ mL min}^{-1}$. Determinations of enantioselectivities were conducted using chiral stationary phase (FS-hydrodex β -TBDAC or β -6-TBDM capillary columns) gas chromatography with a flame ionization detector (cf. Supporting Information for details).

For our ML modeling approach, we combined the reactant and catalyst graph to create a key-intermediate graph (vide supra, Figure 1). The reactant and catalyst graphs were initially created with the RDKit framework⁵⁵ and a graph feature encoder, which was built upon one of DeepChem’s⁵⁶ feature encoders. We used the default settings for the node and edge features. Both individual graphs were combined by locating the node index of boron in the catalyst graph, and the carbonyl oxygen node index of the substrate, and creating a virtual edge between both nodes.

We used Pytorch⁵⁷ and Pytorch Geometric⁵⁸ (PyG) for learning graph representations. Our model consists of a GNN part utilizing a graph attention layer⁵⁹ as implemented in PyG.

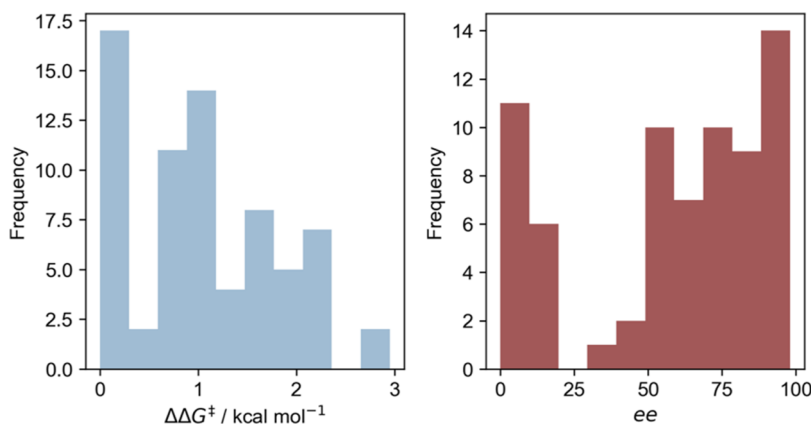


Figure 3. Histograms showing the distribution of our database in $\Delta\Delta G^\ddagger$ (blue) and ee (red) in both domains; the distribution is near-uniform. Such a database is well-suited for ML, in contrast to databases constructed from experimental literature data, which tend to report on the best results only and therefore hamper ML training.

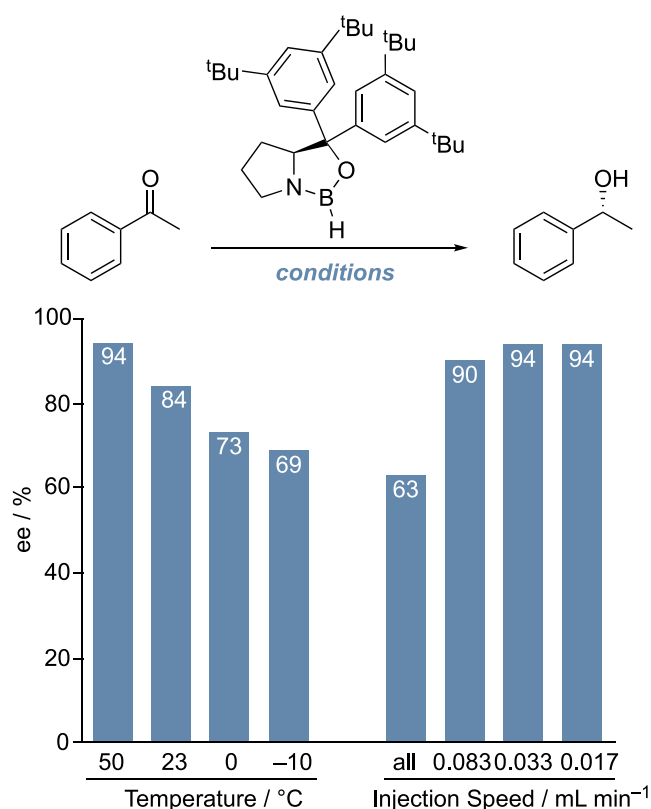


Figure 4. Tests were carried out using acetophenone to establish the reaction conditions at the start of the overall analysis with the shown catalyst. All injection speed tests were carried out at 50 °C.

The node embeddings are then used to generate the molecular embedding via global pooling operations. The molecular embedding is then fed into a feed-forward neural network (FFNN) to predict $\Delta\Delta G^\ddagger$. An overview of our workflow is depicted in Figure 5. Note that only the intermediate before

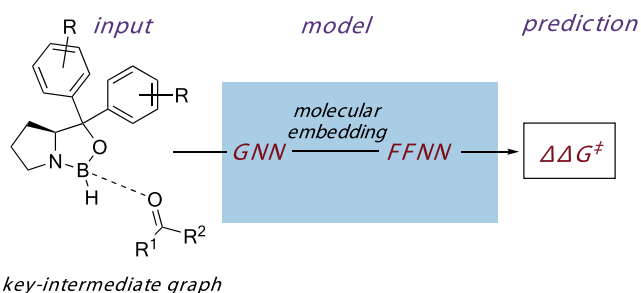


Figure 5. Key-intermediate graph is processed by a GNN that produces node embeddings. Using global pooling operations, molecular embedding is computed and used as input in a feed-forward neural network to predict $\Delta\Delta G^\ddagger$.

the rate determining step is needed (in the form of a molecular graph) to make predictions about the enantioselectivity of the reaction. The hyperparameters of our model were optimized by a Bayesian optimization loop with a Gaussian process as implemented in scikit-optimize.⁶⁰ We optimized the model parameters to minimize the mean squared error (MSE) between predicted and experimental (calculated from gas chromatographic analyses) $\Delta\Delta G^\ddagger$ values. The splitting in training and testing data was performed via a random 80:20 train-test-split. To prevent overfitting, we employed early

stopping. Due to the limited data available, we performed leave-one-out cross-validation (LOOCV) with the scikit-learn⁶¹ package to assess our model's performance reliably and to determine the uncertainty of our model for each data point.

In addition to LOOCV, we checked whether the model "learned the chemistry" behind the CBS reduction by completely excluding a substrate or catalyst from training and then validating. With this approach, we can dissect for which substrates and catalysts the model can accurately predict $\Delta\Delta G^\ddagger$, hinting toward certain strengths and weaknesses of both our model and approach.

Using 100 data points for the LOOCV, we achieved an R^2 -score, root MSE, and mean absolute error (MAE) of 0.95, 0.12, and 0.16 kcal mol⁻¹, respectively, for the combined validation set (Figure 6). The bottom part of the performance

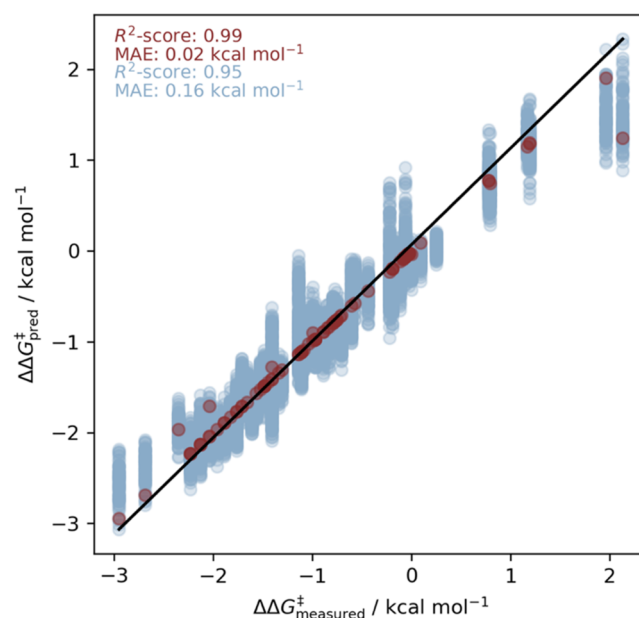


Figure 6. Scatter plot showing the predicted against the experimental $\Delta\Delta G^\ddagger$ values throughout the LOOCV. The black diagonal indicates optimal prediction. Validation points are colored in red, while training points are colored in blue.

plot indicates that the uncertainty is uniformly distributed over the range of possible $\Delta\Delta G^\ddagger$ values, which arguably is due to optimally distributed training data.

The results for excluding substrates and catalysts are shown in the scatter plot depicted in Figure 7. It is apparent that only the two indole-based catalysts (green and gray) are predicted incorrectly when excluded from the training data. In both cases, the model tends to predict values close to the data set's mean. However, excluding catalysts generally results in a smaller deviation from the mean compared to excluding substrates, suggesting the model is more resilient toward missing catalyst data. This observation is evident in the bar plot of Figure 7 and is further supported by the calculated MAEs: substrate exclusion yields a mean MAE of 0.3 kcal mol⁻¹, while catalyst exclusion has a mean MAE of 0.4 kcal mol⁻¹.

Indeed, the model is designed to associate stereochemistry with the sign of $\Delta\Delta G^\ddagger$, given appropriate training data. However, when training data only including examples with $\Delta\Delta G^\ddagger < 0$, it becomes challenging for the model to accurately

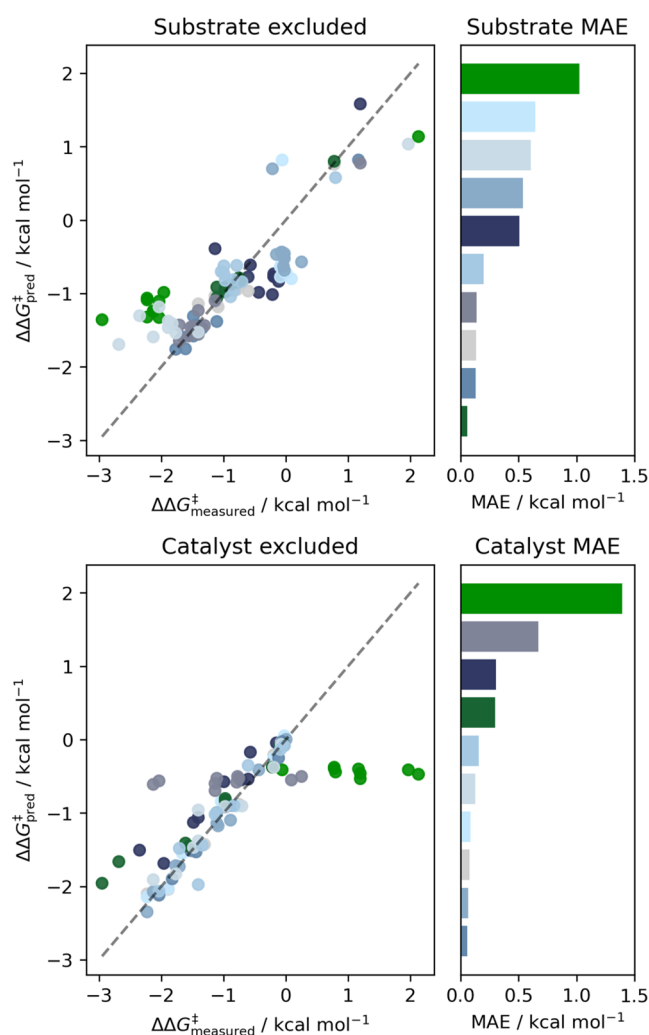


Figure 7. Scatter plots showing the predicted against the measured $\Delta\Delta G^{\ddagger}$ values and bar plots showing the MAE for each excluded molecule (substrate or catalyst) calculated from the scatter plot. This validation method highlights molecules that cause biased predictions, as they exhibit a high MAE.

predict $\Delta\Delta G^{\ddagger} > 0$ values for data points that were not included in the training set, and *vice versa*.

To validate the effectiveness of our ML-prioritized catalyst design approach, we conducted a benchmarking study, which is depicted in Figure 8. We compared the performance of our method against alternative approaches (*vide supra*).

It is clear from Figure 8 that in our benchmark the models based on graphs (purple), and hence GNNs, performed the best with an RMSE $< 0.1 \text{ kcal mol}^{-1}$. This was especially evident when employing our key-intermediate graph instead of the disconnected graph (substrate and catalyst separated). Models based on FFNNs follow closely, working well with Morgan fingerprints of radii 2 and 4. The GBRT and SVR models work better with fingerprints with a radius of 2. The computationally expensive DFT descriptors (ca. 7000 CPU h) are even worse, with only a marginal improvement observed on the computationally cheaper shallow learning methods using fingerprints of radius 4.

We initially let our model predict the selectivity of the catalyst candidates depicted in Scheme 3, for which only the catalyst highlighted by the green disk was predicted to result in

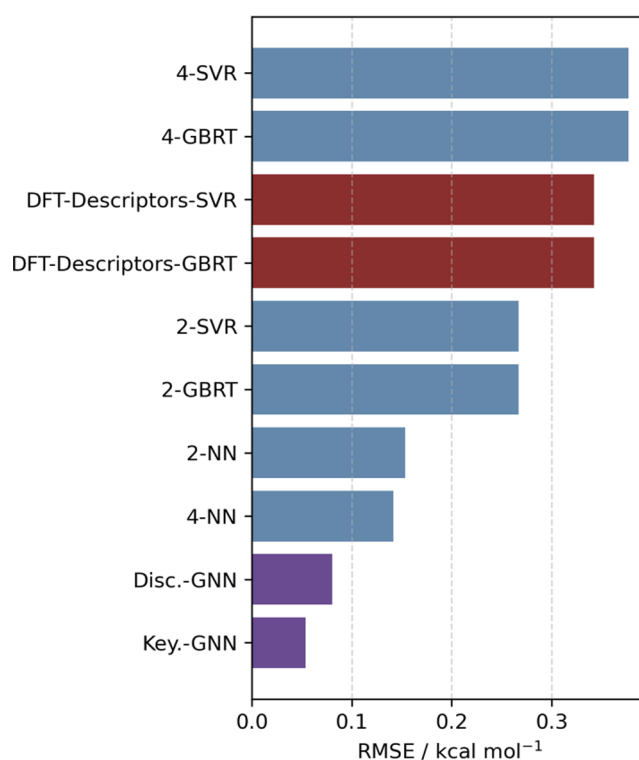


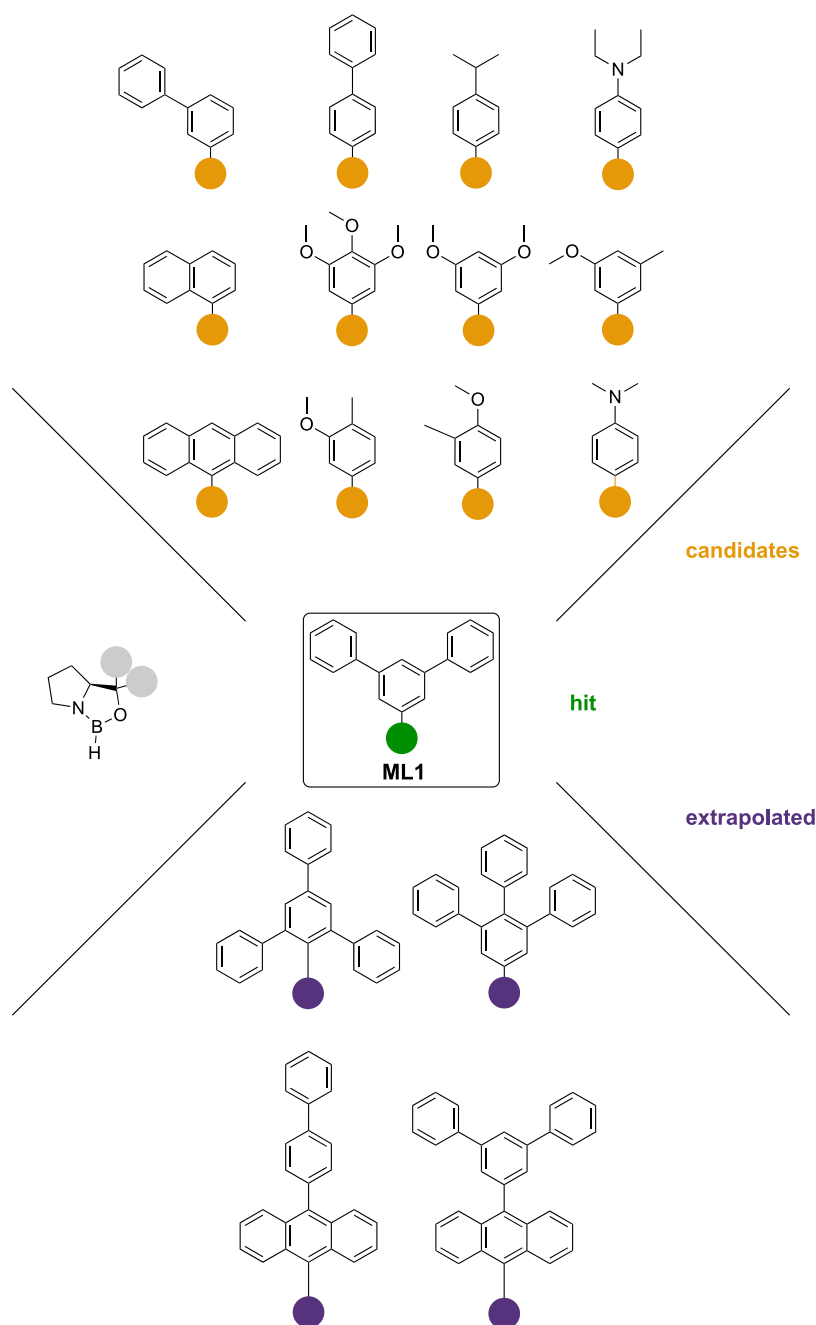
Figure 8. Models' MSE of the validation set for various methods, architectures, and descriptors, sorted in a descending order. The abbreviations used are SVR, support vector machine;⁶² NN, FFNN; and GBRT, gradient boosting regression tree.⁶³ The color indicates the general molecular representation class: blue, Morgan fingerprint;⁶⁴ red, DFT descriptors; and purple, graph-based.

higher selectivity than the best literature-known catalyst for the CBS reduction of butanone.¹

Visualization of feature weights provides insights into the relative importance of each input feature in the model's decision-making process. By examining the magnitude and direction of these weights, it is possible to determine which features have the greatest influence on the model's output and whether the model has learned to focus on relevant information. Techniques such as attention mechanisms or gradient-based saliency maps can be employed to gain insights into the model's decision-making process at the node or edge level.^{59,65–69} These methods allow for the identification of the most influential nodes or edges contributing to the final graph-level prediction, providing a more granular understanding of the model's reasoning. This analysis helps validate that the model has captured useful relationships within the data and does not rely on spurious correlations or irrelevant features. The visualization of feature weights serves as a valuable tool for interpreting the model's behavior and ensuring that it has learned to make predictions based on meaningful patterns in the input data. We focus on a few selected explainable artificial intelligence (xAI) techniques to look at node-feature transformations (visualized with py3Dmol),⁷⁰ node-feature importance, and at relevant walks via GNN layerwise-relevance-propagation (LRP),⁷¹ as shown in Figure 9.

The top part of Figure 9 shows how the location node-feature sum changes after each layer toward the "reactive center". After the first layer, nearly all nodes have a meaningful node-feature sum, while after the last layer, the amplitude of node features is focused on the coordinative bond between the

Scheme 3. Depiction of Our Selection of Catalyst Candidates (Orange Circles), Constructed and Filtered Based on Commercial Availability^a



^aThe catalyst highlighted with the green circle (ML1) was predicted to result in higher selectivity than the best literature-known catalyst. Possible catalyst candidates extrapolated from ML1 are highlighted with purple circles. The gray circles on the L-proline scaffold indicate the place of substitution.¹

carbonyl oxygen and the boron of the catalyst, indicating that the model “learned” to pay attention to the part of the molecule where the chemistry happens. The middle of Figure 9 shows which node features have the most relevance for the models’ predictions. The most important feature is the Cahn–Ingold–Prelog (CIP) code, which describes the stereochemistry of the L-proline-based catalyst and agrees with our expectation that the chirality of the catalysts directly influences the chirality of the product. The second most important feature is having carbon atoms, which obviously trace back to the abundance of carbon atoms in the structures. The third

and fourth places indicate that the model prefers cyclic structures, in particular, aromatic rings. The fifth place again hints at the importance of having a stereogenic center in the catalyst. At the bottom of Figure 9, we showcase the important walks using code-based GNN-LRP.⁷¹ To include node-level visualizations, we added self-loops to the graph. The colors green and purple indicate the positive and negative influences, respectively, toward the models’ prediction on the absolute $\Delta\Delta G^\ddagger$ values. The phenyl rings on the L-proline scaffold are predicted to have a positive influence, which is expected, as an increase in DED groups leads to a higher selectivity.¹ On the

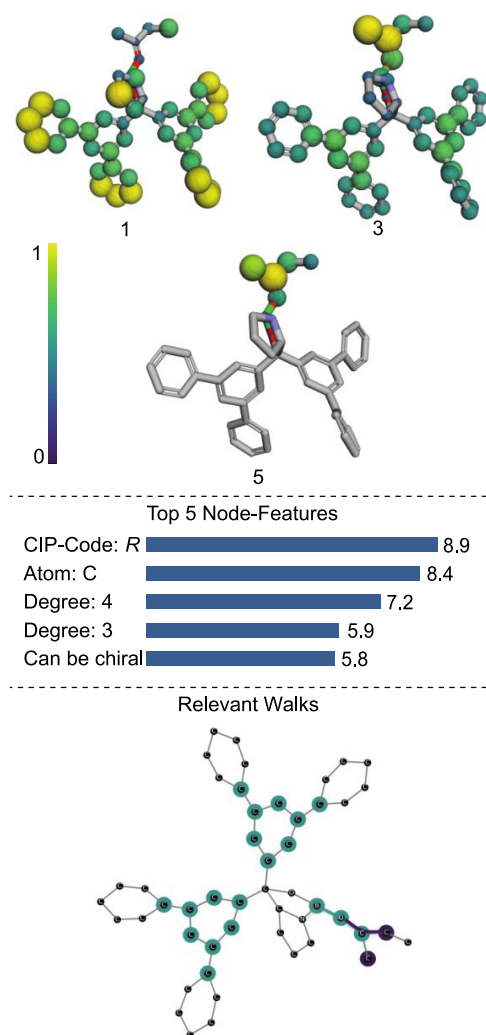


Figure 9. Exemplary explanations for using **ML1** for the reduction of butanone. Top: Visualization of the node-feature sum after layers one, three, and five, as indicators for the importance of the graph structure. The radius scales linearly with the node-feature sum. Middle: Using GNN-Explainer⁶⁵ to look at the top five node-feature importance. Bottom: Using the GNN-LRP approach to look at relevant walks of the model; the colors green and purple indicate positive and negative contributions, respectively. This surrogate model used a graph representation with added self-loops to enable the visualization of node contributions.

other hand, looking at the substrate part of the molecular graph, we see negative influences (highlighted in purple, Figure 9) on the butanone moiety, which aligns with our expectations. This nicely reflects the challenging nature of the butanone reduction (the methyl/ethyl problem) as it generally results in lower enantioselectivity compared to other substrates (vide supra).

Experimental Validation. Using the hitherto unreported **ML1**, we achieved 80% ee (averaged over five repetitions) in the asymmetric reduction of butanone to (*R*)-butane-2-ol. We used **ML1** on all other substrates in our database and extended the database with the new results. One prominent feature of **ML1** is the addition of two phenyl groups to the aromatic ring, which is attached to the L-proline scaffold. Such an extension, i.e., additional aromatic substituents on the phenyl rings that are attached to the L-proline scaffold, has been shown to be

effective, for example, in the enantioselective addition of indole-2-carboxaldehydes to *o*-quinone derivatives.⁷²

The two phenyl rings in the *meta*-position to the L-proline scaffold are excellent DEDs, and the resulting high selectivity of **ML1** in the CBS reduction is in accordance with our previous study and the DED concept.¹ Extrapolating from **ML1** even more DED-rich catalysts can be envisioned (Scheme 2). The downside of the even higher performing catalysts is their challenging preparation, especially in the Grignard reaction for sterically very encumbered substrates.

In the evolving landscape of catalyst design, our work represents a novel integration of ML with traditional synthetic chemistry by utilizing a key-intermediate graph representation, a concept paralleling recent advancements in graph theory applications in chemistry.^{31,39} It is remarkable that the approach otherwise is agnostic to the mechanism of the reaction and that—in using classical thought models—no explicit knowledge about the transition structures for the underlying reactions is required to predict more suitable catalysts.

CONCLUSIONS

We demonstrate that ML can overcome the very challenging “methyl/ethyl problem” in the CBS reduction with an enantioselectivity of 80%. Even a relatively small and therefore “affordable” database with respect to time and effort can leverage the power of ML, provided the experimental data are tightly curated through careful optimization, monitoring, and repetition. The resulting catalyst was an immediate hit, and this suggests that the next step to reach an even better catalyst would be to develop a generative modeling approach or use reinforcement learning.

LOOCV-based analyses show an R^2 -score of 0.95 for prediction accuracy. The achieved RMSE and MAE values of 0.12 and 0.16 kcal mol^{−1}, respectively, demonstrate our method’s high precision and reliability. Moreover, by analyzing the scatter plots, we found that the uncertainty of our predictions is uniformly distributed over the entire range of possible $\Delta\Delta G^\ddagger$ values. This is primarily because our training data are well-distributed and representative. Our approach was validated by including a broad range of substrates and catalysts. We identified a promising catalyst (highlighted in green in Scheme 2) that exhibits higher selectivity than the previously known best catalyst for the CBS reduction of butanone. Additionally, we investigated our models’ predictions via various xAI methods to confirm that our models’ predictions are reasonable.

Our results underline the effectiveness of ML algorithms in catalysis optimization, demonstrating the powerful synergy of synthetic chemistry and ML. Even with a small but high-quality data set, the iterative cycle of synthesis, informed by domain knowledge and synthesis, coupled with ML, can lead to significant advancements. This integrated approach is promising for future development and can complement or even replace traditional design of experiment approaches when high-quality data are available. Future directions to expand on our fundamentally simple key-intermediate graph approach would be the development of a framework that can be universally applied to any catalytic reaction, similar to suggestions by Tran and Ulissi, who proposed that ML and computational methods could accelerate catalyst discovery and optimization.³³

■ ASSOCIATED CONTENT

Data Availability Statement

The code and database used for modeling are also available free of charge at dx.doi.org/10.22029/jlupub-17995.

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.4c01286>.

Experimental; analytic; and modeling details (PDF)

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Notes

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